

EE5750 Quantum Error Correction

Homework 2

Due: 13 Aug

As part of the Homework assessment provide solutions to only the questions marked as **(*Homework*)**. The rest of the questions are practice problems.

Kraus Representation

1. **(*Homework*)** Suppose $\mathcal{E}(\rho) = K\rho K^\dagger$ is the Kraus form of an invertible function $f : \mathcal{X} \rightarrow \mathcal{X}$. What is the Kraus form the inverse function f^{-1} ? Do a sanity check: what is the Kraus form of $f^{-1} \circ f$?
2. **(*Homework*)** *Randomized functions* Consider a randomized algorithm acting on the input alphabet $\mathcal{X} = \{0, \dots, d_1 - 1\}$ with output alphabet $\mathcal{Y} = \{0, \dots, d_2 - 1\}$. The algorithm generates a random output for each possible input with the following conditional probabilities:

$$\mathbb{P}(\text{output} = i \mid \text{input} = j) = p_{i|j}$$

Find a Kraus form for this operation. Verify the truth table (that is, check what is the output state when the input is $|j\rangle\langle j|$), and verify completeness.

3. **(*Homework*)** Find the unitary matrix corresponding to the following invertible functions f . In each case, the function acts on a collection of classical bits, we will denote the individual bits as b_1, b_2 , etc.

- (a) *SWAP gate*: $f(b_1, b_2) = (b_2, b_1)$

Remark: This is the function from Quiz 1. This is called the SWAP gate as it swaps/reorders the two physical bits b_1, b_2 .

- (b) *NOT gate or the X gate*: this gate operates on 1 bit only, output is the logical complement of the input.

Remark: The corresponding unitary is denote as the X operator/matrix.

- (c) *Controlled-NOT or the CNOT gate*: This function works on two bits – one of them is designated as the ‘control’ bit and the other is called the ‘target’ bit. The function flips the value (logical complement) of the target bit if the control bit value is 1, and it copies in the input to the output otherwise.

If the control-bit is b_1 and the target is b_2 , we denote this as $\text{CNOT}_{1,2}$, and we have:

$$\text{CNOT}_{1,2}(b_1, b_2) = (b_1, b_1 \oplus b_2),$$

where $\oplus$ denotes logical XOR operation. Here the control value gets added (XOR-ed) to the target value.

Similarly, if the control is b_2 and target is b_1 , we have

$$\text{CNOT}_{2,1}(b_1, b_2) = (b_1 \oplus b_2, b_2).$$

Find the unitaries corresponding to $\text{CNOT}_{1,2}$ and $\text{CNOT}_{2,1}$.

- (d) *Controlled-CNOT gate or CCNOT or Toffoli gate*: This acts on 3 bits b_1, b_2, b_3 , and it applies $\text{CNOT}_{2,3}$ only when $b_1 = 1$ and acts as an identity function otherwise. So, the value of b_1 controls what happens on the other two bits. Here,

$$f(b_1, b_2, b_3) = (b_1, b_2, b_1 b_2 \oplus b_3),$$

where $b_1 b_2$ is the multiplication operation denoting the logical AND of b_1, b_2 .

4. The Toffoli gate is known to be a universal gate in classical computing, that is, any Boolean function $g(b_1, \dots, b_m)$ can be computed using a circuits consisting of only Toffoli gates and some additional bits (whose values we are allowed to set as per our choice to 0 or 1). These additional bits (which are not inputs to the target function g) are called ‘ancillary’ bits or simply ‘ancilla’.

Can you design circuits based on Toffoli gates for the following classical operations/functions g ? You might have to use ancillary bits in your circuits; if so, identify which bits in your circuits are the function inputs (i.e., b_1, b_2 etc.) and which are the ancillary bits.

- (a) NOT operation
- (b) AND operation
- (c) OR operation

Kronecker Products

5. If A and B are square matrices (not necessarily of the same size) and are both invertible, verify

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}.$$

Remark: Kronecker product of invertible operators is invertible.

6. Consider

$$|a\rangle = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{d_1-1} \end{bmatrix}, |b\rangle = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{d_1-1} \end{bmatrix} \in \mathbb{C}^{d_1} \text{ and } |x\rangle = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{d_2-1} \end{bmatrix}, |y\rangle = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_{d_2-1} \end{bmatrix} \in \mathbb{C}^{d_2}.$$

- (a) Explicitly verify the formula

$$(\langle a| \otimes \langle x|) (|b\rangle \otimes |y\rangle) = \langle a|b\rangle \cdot \langle x|y\rangle.$$

- (b) Verify that $|a\rangle \otimes |x\rangle = 0$ if and only if $|a\rangle = 0$ or $|x\rangle = 0$.

Remark: You might need these results for the next problem.

7. (***Homework***) *Kronecker products and linear independence.*

- (a) Suppose $|\psi_0\rangle, \dots, |\psi_{m-1}\rangle \in \mathbb{C}^m$ are linearly independent. This means that if $\sum_i \alpha_i |\psi_i\rangle = 0$ then $\alpha_i = 0$ for all i .

Now, let $|\phi_0\rangle, \dots, |\phi_{m-1}\rangle \in \mathbb{C}^p$ be any set of m vectors in $\mathbb{C}^p$ (m need not be equal to p , and $|\phi_0\rangle, \dots, |\phi_{m-1}\rangle$ need not have linear independence, these are just any collection of m vectors).

You must prove the following

$$|\phi_0\rangle \otimes |\psi_0\rangle + \dots + |\phi_{m-1}\rangle \otimes |\psi_{m-1}\rangle = 0 \Rightarrow |\phi_i\rangle = 0 \text{ for all } i.$$

Hint: This is one possible technique: find an invertible $mp \times mp$ matrix A such that

$$A(|\phi\rangle \otimes |\psi_i\rangle) = |\phi\rangle \otimes |i\rangle$$

for any $|\phi\rangle \in \mathbb{C}^p$ and for all $i = 0, \dots, m-1$. Once you have such an A , apply the operator A to both sides of

$$|\phi_0\rangle \otimes |\psi_0\rangle + \dots + |\phi_{m-1}\rangle \otimes |\psi_{m-1}\rangle = 0$$

and then make use of the fact that $|0\rangle, \dots, |m-1\rangle$ are the standard basis for $\mathbb{C}^m$.

- (b) Now, we are ready to prove the fact that if $|\psi_0\rangle, \dots, |\psi_{m-1}\rangle$ are linearly independent in $\mathbb{C}^m$, and if $|\zeta_0\rangle, \dots, |\zeta_{p-1}\rangle$ are linearly independent in $\mathbb{C}^p$, then the collection of mp vectors $\{|\zeta_i\rangle \otimes |\psi_j\rangle : i = 0, \dots, p-1, j = 0, \dots, m-1\}$ is linearly independent in $\mathbb{C}^{mp}$.

Prove this using the result of part (a).

Hint: Start from $\sum_i \sum_j \alpha_{i,j} |\zeta_i\rangle \otimes |\psi_j\rangle = 0$, and reduce this to the equation considered in part (a); use the linearity of Kronecker products to do this. Finally, show that $\alpha_{i,j} = 0$ for all i, j .

Remark: We saw in Lecture 4 that if $\{|\zeta_i\rangle\}$ and $\{|\psi_j\rangle\}$ are orthonormal then so are $\{|\zeta_i\rangle \otimes |\psi_j\rangle\}$. The current problem shows that a similar result holds for linear independence too. In fact, since there are precisely mp vectors in $\{|\zeta_i\rangle \otimes |\psi_j\rangle\}$ we see that this collection is a basis for $\mathbb{C}^{mp}$.

This result extends to Kronecker products of matrices as well. We will see an example (applied to matrices) in the next question.

8. (***Homework***) For simplicity and concreteness we will consider Kronecker products of 2×2 matrices only, i.e., both A and B in $A \otimes B$ are 2×2 . Note that $A \otimes B$ is a 4×4 matrix.

- (a) Find 4 matrices $M_1, \dots, M_4$ in $\mathbb{C}^{2 \times 2}$ (the space of 2×2 complex matrices) such that every matrix in $\mathbb{C}^{2 \times 2}$ can be expressed as linear combination of $M_1, \dots, M_4$.

Hint: Although there are many solutions, you can save a lot of work by a suitable choice of $M_1, \dots, M_4$ that can be easily expressed in terms of the standard basis $|0\rangle$ and $|1\rangle$.

- (b) Now verify that any 4×4 matrix can be written as a linear combination of the 16 matrices:

$$M_i \otimes M_j, \text{ where } i, j = 1, 2, 3, 4.$$

Remark: In general if $\{M_i : i = 1, \dots, m^2\}$ is a basis for $\mathbb{C}^{m \times m}$ and $\{N_j : j = 1, \dots, n^2\}$ is a basis for $\mathbb{C}^{n \times n}$, then

$$\{M_i \otimes N_j : i = 1, \dots, m^2, j = 1, \dots, n^2\}$$

is a basis for $\mathbb{C}^{mn \times mn}$. In particular, every $mn \times mn$ matrix A can be expressed as linear combination of Kronecker products

$$A = \sum_i \sum_j \alpha_{i,j} M_i \otimes N_j.$$

This is an **important observation**.